



The role of transport sector integration in achieving high shares of renewable energy in African islands: The case of Santiago, Cabo Verde

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ABSTRACT

Renewable energy sources offer the opportunity to decarbonize both the power and transport sectors through the adoption of electric vehicles, especially in resource-rich islands, such as those in Africa. However, the role of electric mobility, particularly vehicle-to-grid technology, as a system-balancing solution remains underexplored for these islands. This study addresses this gap by using the EnergyPLAN model with hourly resolution to assess the integration of electric vehicles, smart charging, and vehicle-to-grid services into a high-renewable energy system on Santiago Island, Cabo Verde. The analysis is based on scenario simulations for 2030 and 2050, assessing the impact on system reliability, carbon dioxide emissions, and system costs. Results demonstrate that vehicle-to-grid services can significantly increase the share of renewable energy penetration, with electric vehicles' batteries providing short-term system balancing and storage services as early as 2030. By 2050, a fully renewable power system with a high share of electric vehicles is technically and economically feasible. However, it requires coordinated planning between the power and transport sectors. The insights from this study highlight the potential for Santiago Island to serve as a scalable model for electric mobility, smart and integrated energy systems in the West African region, offering insights for other Small Islands Developing States aiming for energy self-sufficiency, affordability, and sustainability.

Abbreviations

A	Amperes
ASIF	Activity Structure Intensity and Fuel Composition
BAU	Business as Usual
BEVs	Battery Electric Vehicles
CAPEX	Capital Expenditure
CEEP	Critical Excess Electricity Production
CO ₂	Carbon Dioxide
ESS	Energy Storage Systems
EVs	Electric Vehicles
GHG	Greenhouse Gas
HDDVs	Heavy-Duty-Vehicles
ICEVs	Internal Combustion Engine Vehicles
LDVs	Light-Duty-Vehicles
MSW	Municipal Solid Waste
OPEX	Operational Expenditure
PHS	Pumped Hydro Storage
PP	Power Plants
PV	Photovoltaics
RE	Renewable Energy

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REPS	Renewable Energy Power Sector
RES	Reference Energy System
SIDS	Small Island Developing States
SM	Smart Mobility
TDES	Total Electricity Supply
VAT	Value Added Tax
V2G	Vehicle-to-Grid

1. Introduction

Transportation is a major driver of fossil fuel consumption and greenhouse gas (GHG) emissions on many islands [1]. The sector's heavy reliance on imported fossil fuels not only increases energy costs but also exposes islands to oil price volatility and supply chain disruptions. Small Island Developing States (SIDS) often face some of the highest energy prices globally [2], with oil costs reaching 3–4 times those of mainland settings [3]. Moreover, SIDS face unique

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environmental and economic challenges, including land scarcity, reliance on external resources, isolated power systems, and high vulnerability to the impacts of climate change [4]. Given these constraints, a comprehensive approach to energy planning that integrates the transportation sector into the broader energy transition is essential for achieving energy security and fostering sustainable development.

Transport sector electrification is recognized as a key strategy for the SIDS energy transition [2]. The adoption of battery electric vehicles (BEVs), combined with smart charging and vehicle-to-grid (V2G) services, has been extensively studied as a means of increasing renewable energy (RE) integration, reducing fossil fuel dependence, and lowering overall system costs (e.g., Dorotić et al. [5], Colmenar-Santos et al. [6], Zhang et al. [7]).

Smart charging, sometimes referred to as unilateral V2G [8], allows flexible charging management, aligning the charging of electric vehicles (EVs) with periods of high renewable generation or low electricity demand, thereby supporting grid stability [9]. V2G technology goes a step further by enabling bi-directional energy flows, allowing BEVs to function as distributed storage units, storing electricity when there is a surplus and feeding back into the grid during times of low renewable generation [6]. These capabilities enhance the flexibility of energy systems that rely on intermittent RE sources.

1.1. Existing research on island energy systems

A substantial body of literature research supports the technical and economic feasibility of fully RE-based systems in island settings. A recent comprehensive literature review analysed 81 high-quality articles, including real-world case studies of islands transitioning to 100 % RE systems, providing valuable insights into the technologies, strategies, and policies that enable such transitions [10]. Most studies have found and concluded that high-RE scenarios are “technically feasible and economically viable” when leveraging local renewable resources combined with Energy Storage System (ESS) and demand response strategies.

For example, Arévalo et al. [11] conducted a long-term feasibility study of the Galapagos Islands using EnergyPLAN and artificial neural network models, showing that a fully renewable system by 2050 is both technically viable and economically beneficial when integrating photovoltaics (PV), wind, and storage with high demand forecasting accuracy. Similarly, Graça Gomes et al. [12] found, through a novel optimization model, that a decarbonized and smart energy community on Corvo Island (Portugal) is achievable when implementing onshore technologies coupled with hydrogen-powered heating, while also demonstrating that a system based on solar thermal heating offers the most cost-effective decarbonization pathways.

Several studies have also explored how integrating EVs and V2G services within island energy systems can increase RE penetration and reduce fossil fuel dependency, thereby improving the technical feasibility of 100 % RE systems. For instance, Marcinkowski & Barros [13] modelled a 100 % RE system scenario for Madeira (Portugal) and showed that integrating V2G and other smart technologies could reduce biomass use by 50 % and eliminate curtailment. Child et al. [14] demonstrated how BEVs and electrofuels could support a low-carbon transport pathway for Åland (Finland), identifying V2G and power-to-gas as key storage solutions. Alves et al. [15] demonstrated how interconnecting Faial and Pico’s (Portugal) energy systems with integrated V2G services could eliminate fossil fuel dependency. Further, Colmenar-Santos et al. [16] investigated how plug-in vehicles could contribute to demand peak-shaving as well as possible cost reductions for EV owners and the system operator. The authors estimated that vehicle-to-home applications could lower energy costs by 50 % and flatten peak demand while improving RE integration.

Despite these advancements, most existing research focuses mainly on European and Pacific islands. For example, Meschede et al. [17] reviewed 97 papers but found only four studies on African SIDS,

revealing a significant regional research gap. In particular, the integration of the transport sector into 100 % RE power system studies for African SIDS remains limited. The African SIDS, which include Cabo Verde, Comoros, Guinea-Bissau, Mauritius, São Tomé and Príncipe, and Seychelles, have been the subject of various 100 % RE planning studies, but few have considered how EVs and V2G services could support this transition. For example, research conducted for Cabo Verde and Mauritius did not consider transport sector integration (e.g., Nordman et al. [18], Khoodaruth et al. [19], Ferreira et al. [20]). While some recent efforts have included sector coupling (e.g., Pombo et al. [21]), a comprehensive analysis of transport integration is missing, especially an assessment of how EVs and V2G services can play a role in achieving higher shares of RE. Table 1 provides an overview of existing studies on 100 % RE planning for African SIDS, illustrating the lack of research on transport sector integration.

Moreover, most of these studies do not consider a complete hourly-resolution model, rather use monthly averages, representative weeks, or time slices, which makes it difficult to accurately capture the dynamic interactions between variable renewables, smart charging strategies, and V2G interactions with the grid. Addressing this methodological gap is crucial for understanding the real potential of EVs and V2G to support high RE shares in island systems. This research gap is particularly relevant as EV adoption is accelerating globally in both passenger and commercial markets [25] and is increasingly recognized as a key enabler of the energy transition in Africa [22,23]. However, the exclusion of African regions from mainstream EV research could hinder the transformation of their transport sectors [26]. Similar to how Madeira (Portugal), Orkney (United Kingdom), and Samsø (Denmark) have been used as pilot sites for testing smart energy technologies [27], African SIDS present a unique opportunity to serve as testbeds for innovative smart charging and V2G strategies to demonstrate the feasibility of integrated power and transport solutions, potentially informing broader energy transitions across mainland countries.

1.2. Paper contribution and structure

This study, therefore, addresses the identified research gap by assessing the role of EVs and V2G services in facilitating high RE integration in an African island context using a detailed hourly resolution

Table 1

Overview of existing studies focused on 100 % RE planning for African SIDS.

Reference	Island	Methodology/ Software	Sectors included	Time resolution
Nordman et al. [18]	Cabo Verde	Systematic Review	Power	N/A
Khoodaruth et al. [19]	Mauritius	Systematic Review	Power	N/A
Ferreira et al. [20]	Santiago (Cabo Verde)	Generation expansion planning	Power	Monthly
Pombo et al. [22]	São Vicente (Cabo Verde)	Generation expansion planning	Power	Hourly, but representative 12-weeks per year
Timmons et al. [23]	Mauritius	OSeMOSYS	Power	Hourly, but limited to 730 time slices per year
Timmons et al. [24]	Mauritius	OSeMOSYS	Power	Hourly, but limited to 730 time slices per year
Pombo et al. [21]	Santiago (Cabo Verde)	Flexibility generation expansion planning	Power, desalinization, hydrogen and transport (excluding V2G services)	Hourly, but representative 12-weeks per year

energy system modelling. Unlike the previous studies on 100 % RE for African SIDS that focus mainly on the power sector or consider a single, aggregated mobility demand, this research represents the road-transport fleet by vehicle class, stock evolution, driving patterns, and battery characteristics for every hour of the year. This granularity makes it possible to co-optimize smart-charging schedules and bidirectional V2G dispatch in real time, providing a sophisticated, coupled power-and-transport simulation that captures the flexibility potential of transport electrification.

Specifically, this research evaluates how EVs and V2G technologies can serve as primary ESS, reducing reliance on fossil fuel-based flexibility options. Santiago Island (Cabo Verde) is selected as the case study due to data availability and the government's ambitious target to phase out internal combustion engine vehicles (ICEVs) by 2050 alongside the power sector transition to RE, hence making it an ideal testing ground to examine the role of transport sector integration in achieving high shares of RE.

The key contributions of this research are as follows.

- First high-resolution modelling of EV and V2G integration in an African SIDS using EnergyPLAN: This study provides the first hourly-resolved analysis of EV integration and V2G-based flexibility in an African island, providing an in-depth technical, economic, and environmental assessment of EV adoption, smart charging, and V2G services in an African island context.
- Assessment of short and long-term policy implications: The study assesses the short-term (2030) and long-term (2050) impacts of government decarbonization policies, offering insights into necessary policy adjustments.
- Fostering discussion on EV adoption in Africa: To date, no previous studies have focused on transitioning to a fully RE system for the region and explicitly examined the systemic impact of direct electrification on the power sector. Thus, this research contributes to filling this gap by evaluating how smart charging and V2G technologies reshape electricity demand and storage needs. In doing so, this research aims to stimulate discussion on the opportunities and challenges of EV adoption in African islands, providing valuable insights for sub-Saharan countries considering similar transitions.

The remainder of this paper is structured as follows: Section 2 details the methods and modelling approach; Section 3 presents the results and sensitivity analyses; Section 4 discusses the findings; and Section 5 concludes by summarising key takeaways, policy implications, and

directions for future research.

2. Methods and data

This section details the methodology and data sources used. The overall research framework is shown in Fig. 1 and consists of three main steps.

- Data Collection and Forecast: Collecting input data on energy demand and supply for electricity and transport sectors, followed by bottom-up calculations to estimate future consumption levels.
- Energy System Modelling: Modelling energy transition scenarios using EnergyPLAN to simulate the integration of RE, EVs, and V2G operations.
- Scenario Analysis and Evaluation: Analysing key performance metrics such as diversification of the generation mix, RE penetration, EV charging strategies, emissions, system costs, and the impact of V2G integration on system flexibility.

Each of these steps is explained in detail in the subsequent subsections.

2.1. Reference energy system

The reference energy system (RES) for Santiago Island is simulated using EnergyPLAN, with 2021 as the base year. Santiago is the largest island in the Cabo Verde archipelago, and it is home to half of the country's population, consuming over 50 % of the nation's total energy. Input data used for the simulation are electricity generation and demand [28], hourly renewable output [29], fuel consumption [30], and transport sector statistics [28,29]. Because national reports often aggregate data at the country level, Santiago-specific energy estimates were derived using local expert datasets and disaggregated consumption patterns [31].

In 2021, Santiago's electricity system relied heavily on imported fossil fuels, with 84 % of generation coming from oil. Installed capacity included 66.1 MW from thermal plants, 9.4 MW from wind, and 4.3 MW from solar PV. Peak demand reached 40.67 MW in October 2021, and total electricity consumption stood at 252 GWh that year [28].

This fossil fuel dependency is also reflected in the transport sector, accounting for 42 % of the island's total oil consumption [31]. Of 50,650 registered vehicles, 73 % were light-duty vehicles (LDVs), 15 % were heavy-duty vehicles (HDVs), and 12 % were motorcycles and trailers

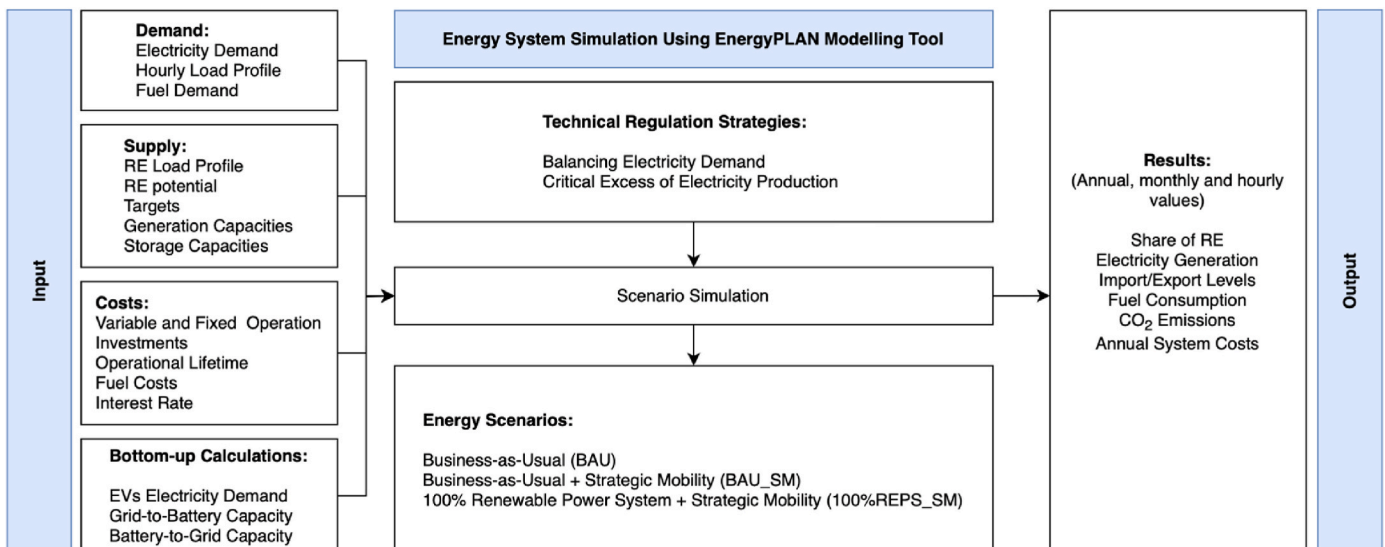


Fig. 1. Overview of the methodological approach.

[32]. EVs were nearly absent, making up just 0.6 % of the LDV fleet. Table 2 presents the fuel distribution of the vehicle stock.

EnergyPLAN model outputs were validated by comparing simulated results to official energy statistics. Table 3 presents the comparison between modelled values and actual data from 2021, demonstrating the model's accuracy in replicating the RES.

2.2. Energy demand forecasting

Electricity demand comprises consumption across all sectors, including industry, heating, cooling, and desalination units, transmission and distribution losses, and is aggregated into a unified demand profile. The forecast is based on Cabo Verde's Electricity Master Plan 2019–2040 [34]. According to the Master Plan, Santiago is projected to account for 40 % of the national electricity demand by 2030, increasing to 42 % by 2040. For the period beyond 2040, electricity demand is projected using a linear extrapolation method, assuming a continued trend consistent with historical growth. No additional structural shifts in demand—such as those resulting from rapid electrification of transport or industry—are assumed beyond the current policy horizon. The forecasted demand is held constant across all modelled scenarios.

For the transport sector, this study employs the IEA Mobility Model [35] based on the ASIF (Activity, Structure, Intensity, and Fuel Composition) framework to estimate the sector's energy demands. This model approach, extensively used in other studies (e.g., Arioli et al. [36], Deendarlianto et al. [37], Siregar & Möller [38], Yuan et al. [39]), allows for structured projections of travel demand based on vehicle stocks and fuel consumption as per equation (1):

$$ED_f = \sum_v VS_v \times VP_{v,f} \times FE_{v,f} \times TD_v \quad (1)$$

Where.

- ED_f : annual energy demand of each fuel type f
- f : fuel type;
- v : vehicle type;
- VS_v : the vehicle stock of each type v ;
- $VP_{v,f}$: the share of vehicle type v using fuel type f ;
- $FE_{v,f}$ the fuel economy of vehicle type v using fuel type f ;
- TD_v : the annual distance travelled.

Vehicles are categorised as motorcycles, cars, buses, minibuses (vans), pickups, trucks, and other HDVs for agriculture and construction, based on the data from the national road transport regulator, 'Direção Geral dos Transportes Rodoviários' (DGTR) [28,29], and data from the Alternative Fuels Data Centre [40]. Forecasts for vehicle stocks were calculated based on historical trends and predicted population growth for the island obtained from Ref. [41]. Possible changes that could result in vehicle stock reduction over the year were not considered in this study. The key assumptions considered regarding fuel consumption and fuel economy across vehicle categories can be found in Table 4.

Furthermore, it is assumed that electric equivalents will eventually replace these vehicles. The technical specifications of selected EVs considered in the simulations are detailed in the Supplementary Material (Table 2–A). Larger HDVs, such as construction and agriculture vehicles, are included in all scenarios to reflect their fuel use and emissions. However, they are not considered for electrification due to

Table 2
Distribution of vehicles by category and fuel type in 2021 [33].

Vehicle category	Diesel	Gasoline	Electricity
HDV (includes trucks, buses, minivans, construction and agriculture type of vehicles)	99.0 %	1.0 %	–
LDV (passenger cars, pickups, others)	55.7 %	43.7 %	0.6 %
Motorcycles and trailers	–	100.0 %	–

Table 3

Validation of reference energy system compared with the national statistics.

	2021 Reference Energy System	EnergyPLAN
RE electricity generation (GWh/year)	40.32 [28]	40.44
Thermal electricity generation (GWh/year)	211.99 [28]	211.87
Share of renewable electricity production (%)	16 [28]	16
Total oil consumption (GWh/year)	1953 [30]	1951.5

data limitations and the technical challenges of direct electrification [42]. Nonetheless, their diesel and gasoline consumption are modelled without estimating the electricity that would be required to replace it. Finally, it is also assumed that the existing grid infrastructure can accommodate the added demand and flexibility from V2G operations; thus, this study does not explicitly model potential grid upgrade requirements.

2.3. Energy system modelling tool - EnergyPLAN

The integration of EVs, V2G, and RE is simulated using EnergyPLAN (version 16.22), a well-established energy system analysis tool developed by the Sustainable Energy Planning Research Group at Aalborg University [43]. The model was selected because it simulates hourly system operation, capturing renewable intermittency and demand fluctuations; it supports sector coupling by integrating electricity, transport, and storage systems and provides technical and economic assessments of system costs, emissions, and energy flows. In this study, EnergyPLAN operates under a “Technical Simulation Strategy”.

To integrate V2G services, additional key parameters were defined in EnergyPLAN, including BEV battery capacity and efficiency; grid-to-battery (i.e., charge capacity) and battery-to-grid (discharge capacity) limits. Additionally, hourly EV availability for grid interaction and smart charging participation levels were specified. These inputs are modelled using an aggregated approach, where, e.g., the total battery capacity of all EVs is represented as a single dynamic energy storage unit. The availability of BEVs for grid connection is defined by their hourly distribution, which reflects the driving patterns and charging behaviours of the vehicle fleet [43]. This distribution specifies the share of vehicles driving at a given moment, which are unavailable for grid services, and the remaining parked vehicles, which can participate in V2G or smart charging, depending on user preferences and charging infrastructure accessibility. EnergyPLAN's user manual offers a step-by-step explanation for calculating these parameters. Comparable modelling approaches in island settings have been explored in Refs. [12,35]. Inputs for smart charging and V2G modelling are provided in the Supplementary Material (Table 2–B and Fig. 2-A and 2-B).

2.4. Scenarios definition

In this research, three energy transition scenarios are modelled.

- Business-as-Usual (BAU): Follows national RE plans—targets 50 % penetration by 2030 and 100 % by 2050—but EVs do not engage in smart charging.
- BAU with Strategic Mobility (BAU_SM): Introduces smart charging and V2G services (50 % BEV participation).
- 100 % RE Power System (100 %REPS_SM): Maximizes RE capacity and assumes 75 % V2G participation.

A summary of the key input parameters for each scenario is provided in Table 5.

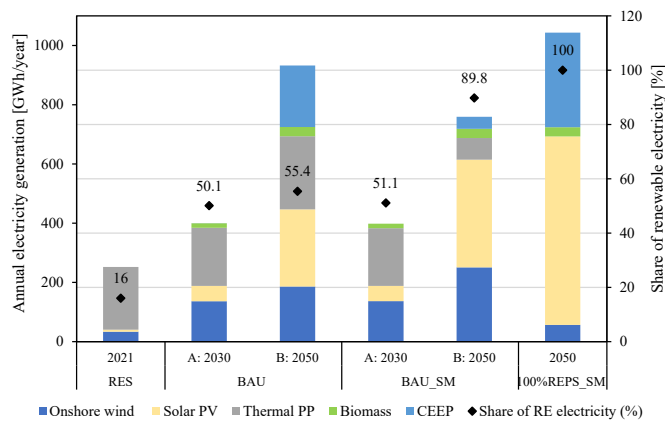
2.4.1. Business-as-usual (BAU) – 2030 and 2050

The BAU scenario models Santiago Island's energy system under

Table 4

Annual transport demand statistics and number of vehicles per target year.

Vehicle Type	Number of Vehicles ^a			Share of vehicles consuming diesel ^b	Share of vehicles consuming gasoline ^b	Fuel Economy Diesel (km/l) ^c	Fuel Economy Gasoline (km/l) ^c
Year	RES	2030	2050	RES			
Cars	33926	41791	63203	50 %	50 %	11.7	10.4
Motorcycles	5922	7295	10867	0 %	100 %	21.1	18.7
Transit Buses	150	185	275	100 %	0 %	1.6	1.4
Minibuses (van)	300	370	550	100 %	0 %	8.5	7.6
Pick-ups	3000	3696	4553	95 %	5 %	8.5	7.6
Trucks	3841	4732	7049	98 %	2 %	2.7	2.4
HDVs Agriculture	390	481	716	100 %	0 %	1.1	1.1
HDVs	3121	3845	5727	100 %	0 %	2.3	2.2
Construction							

^a Based on data from Ref. [32]. “Share of vehicle per category and islands.”^b Based on data from Ref. [33]. “Share of fuel consumption by vehicle category.”^c Based on data from Ref. [40]. “Average fuel economy of Major vehicle categories.”**Fig. 2.** Annual electricity generation and share of RE-generated electricity across all scenarios.

current national policies, incorporating targets from the “Electricity Master Plan 2018–2040” [34] and the “Electric Mobility Letter” [44]. It evaluates whether existing RE expansion plans are sufficient to meet national goals of 50 % RE by 2030 and 100 % RE by 2050, which remain consistent across all scenarios. RE capacities are constrained to priority projects as identified in Ref. [45]. The EV penetration rates, aligned with national targets, are set at 10 % by 2030 and 85 % by 2050, with charging behaviour assumed to follow ‘dump charging’. This behaviour assumes that EVs charge “uncontrolled” after returning home in the evening [46]. A summary of the main strategies adopted from national reports can be seen in the [Supplementary Material Table 1–A](#).

Table 5

Key input parameters for all scenarios.

Scenarios	Year	El. Demand Power sector [GWh]	RE Gen. Capacity [MW]			Share of BEVs	El. Demand Transport Sector [GWh]		Thermal Gen. Capacity [MW]	PHS ^a Capacity [MWh]	V2G Storage Capacity [MWh]
			Wind	Solar	MSW		Dump Charge	Smart Charge			
RES	2021	252.3	9.35	4.28	–	–	–	–	66.1	–	–
BAU	A: 2030	337.2	33.0	30.0	15.2	10 %	15.49	–	66.1	120	–
	B: 2050	423.3	76.5	243.0	30.4	85 %	152.24	–	65.5	600	–
BAU_SM	A: 2030	337.2	33.0	30.0	15.2	10 %	1.49	14	53.9	–	239
	B: 2050	423.3	76.5	243.0	30.4	85 %	17.54	134.67	43.0	600	2429
100 % REPS_SM	2050	423.3	9.0	480.0	30.4	85 %	17.54	134.67	–	600	2429

^a Pumped Hydro Storage.

2.4.2. BAU with Strategic Mobility (BAU_{SM}) – 2030 and 2050

The BAU_{SM} scenario builds on the BAU scenario by introducing smart charging and V2G services to improve RE integration and grid flexibility while reducing fossil fuel reliance.

RE capacity remains unchanged from BAU, however, flexibility mechanisms are reconfigured to prioritize EVs as the main source of storage and system flexibility. This operational prioritization makes sense from a system perspective, as V2G batteries are distributed, co-located with demand, and highly responsive for daily balancing, which helps reduce grid losses and overall system costs [47]. Smart charging enables EVs to charge in response to RE availability, avoiding peak demand. V2G participation is implemented for 50 % of private cars, buses, and minivans, enabling bidirectional energy flow to support grid stability, with the remaining EVs participating in conventional dump charging. These configurations are also applied in the 100 % REPS_{SM} scenario.

The primary objective of this scenario is to assess how controlled EV charging and V2G impact grid performance by comparing system flexibility, emissions reduction, and fuel savings against the BAU case, with BEVs serving as the primary ESS. Finally, a sensitivity analysis also examines the effects of implementing smart charging without V2G capabilities, assessing if and how controlled charging alone can support grid stability, fuel savings, and RE integration.

2.4.3. 100 % renewable energy power system with Strategic Mobility (100 %REPS_{SM}) – 2050

This scenario represents the most ambitious pathway, examining the technical feasibility of a 100 % RE-based power system while accommodating transportation electrification targets.

In the power sector, all fossil-based dispatchable capacity is

removed, creating a system that relies entirely on renewable sources. RE deployment is not limited to priority projects but instead scaled to Santiago's full RE potential [48, p. 50].

BEVs function as the primary ESS thanks to V2G services. Smart charging is fully optimized, ensuring EV demand aligns with RE supply. Lastly, a sensitivity analysis examines how increasing V2G participation from the modelled 50 %–75 % affects grid flexibility in a fully renewable-based power system.

2.5. Costs estimation

There is no clear consensus in the literature on how to account for the costs associated with transitioning from ICEVs to BEVs in energy system models. Approaches vary depending on study scope and modelling objectives: For instance, Child et al. [49] included the full cost of vehicles, considering the changing BEV-to-ICEV ratio across different scenarios. Siregar & Möller [38] performed the economic assessment of vehicles by including only the price difference between ICEVs and BEVs. Cantarero [50] accounted for investment in charging infrastructure for an electric bus system but excluded the cost of light-duty BEVs, considering these as private investments. However, the study included battery costs for V2G services, recognizing their role in grid flexibility (e.g., load balancing and frequency regulation). In a similar approach, Child & Breyer [51] modelled only the cost of BEV batteries, highlighting their importance as grid storage assets rather than personal vehicles.

In line with the methodologies of Cantarero and Child & Breyer, this study excludes the full purchase cost of BEVs and considers only the additional battery costs for vehicles participating in V2G services, assuming that private consumers bear vehicle acquisition costs. Grid infrastructure upgrade costs are also excluded, as Santiago's existing transmission and distribution system is expected to accommodate the projected EV penetration levels without significant reinforcement. This is justified by Cabo Verde's national EV promotion program, supported by the NAMA Facility, which includes planned investments in public charging infrastructure and grid readiness [52]. All assumed cost data, including capital expenditures (CAPEX), operation and maintenance costs (OPEX), and technology lifetimes, can be found in Table 3–A of the Supplementary Material. These assumptions are aligned with market projections for RE technologies, ESS, and V2G-compatible batteries.

3. Results

In this section, the results from the modelled scenarios for Santiago Island described in section 2 are presented. Key insights are drawn for both the 2030 and 2050-time horizons, with a particular focus on the impact of EV smart charging and V2G services in supporting a highly renewable power system, considering EnergyPLAN's hourly resolution.

3.1. Electricity and transport demand

The projected annual electricity demand is estimated to reach 337.2 GWh by 2030 and 423.28 GWh by 2050, driven by population growth, industrial demand, and transport electrification. With 10 % fleet electrification by 2030, oil savings total 85.7 GWh/year. By 2050, with 85 % fleet electrification, gasoline and diesel replacement yields 812.7 GWh in oil savings.

The resulting electricity demand for transport is estimated at 15.5 GWh/year in 2030 and 152.2 GWh/year in 2050. This represents 4 %–20 % of the total electricity consumption in the *BAU* scenarios and 20 %–48 % in the *BAU_{SM}* scenarios. In the *100 % REPS_{SM}* scenario, transport electricity demand accounts for 48 % of total electricity demand, highlighting the significant impact of vehicle electrification on the energy system.

3.2. Electricity supply

Fig. 2 presents the total electricity supply (TES) across all scenarios, showing a significant increase in electricity generation by 2050 to accommodate growing EV adoption and higher RE shares.

In the *BAU* scenarios, the TES rises from 252 GWh in 2021 to 400 GWh in 2030 and 933 GWh in 2050. Fossil fuels-based thermal PP remain a significant contributor, providing 49 % (196 GWh) in 2030 and 26 % (247 GWh) in 2050. The maximum installed solar PV capacity (243 MW by 2050) contributes 41 % of TES with consistent daytime generation, while wind and biomass (waste) supply the remaining 33 % of renewable generation.

In contrast, *BAU_{SM}* and *100 % REPS_{SM}* scenarios with V2G storage enhance the system flexibility during periods of low availability. In the *BAU_{SM}* scenarios, the TES reaches 398 GWh and 759 GWh in 2030 and 2050, respectively, with fossil fuel reliance reduced to just 10 % by 2050. Critical excess electricity production (CEEP) is fully eliminated by 2030 and reduced to 5 % in 2050. In the *100 % REPS_{SM}* scenario, TES increases to 1044 GWh due to the need for overcapacity in renewables to meet electricity demand at all hours. Solar PV dominates the energy mix, providing 90 % of the TES, while wind power complements it primarily by supporting overnight EV charging.

Fig. 3 highlights the hourly imbalance between electricity generation and demand during three representative days in September in the *BAU* 2050 scenario. This period corresponds to the island's rainy season (July to October) when wind output decreases and solar generation is lower due to cloud cover (see Fig. 4-A in Supplementary Material). As a result, the system requires an additional 65.5 MW of thermal capacity to meet the demand. This highlights the system's inflexibility in the absence of efficient seasonal storage to shift the excess RE generation to periods of low supply.

3.3. The role of energy storage solutions

ESS plays a crucial role in mitigating the fluctuations in RE generation, addressing supply gaps, and ensuring demand is met during periods of low generation. In Cabo Verde's strategic energy planning, PHS has been identified as a key ESS for Santiago Island, and it is set as the main ESS in the *BAU* scenario, considering capacities pre-established in the Electricity Master Plan.

Fig. 4 illustrates the annual discharge load provided by the ESS across all scenarios. The *BAU* scenarios rely exclusively on PHS and present the lowest storage utilization, as thermal PP continues to provide much of the balancing capacity. The annual discharge output from PHS increased from 7 GWh in 2030 to 137 GWh in 2050, yet remains insufficient to prevent curtailment (22 % of TEP) during high RE generation periods, as demonstrated in Section 3.2. The limited operational role of PHS is consistent with other island systems (e.g., Kyriakidis et al. [53]) and aligns with the results from Pombo et al. [21] for Santiago's future energy planning, where initial RE integration stages often rely on dispatchable generation rather than storage for flexibility. Consequently, RE penetration, although reaching the policy targets of 50 % by 2030, only increases marginally to 55 % by 2050, despite substantial capacity expansion.

In the *BAU_{SM}* scenarios, V2G-enabled BEVs fundamentally change the operational dynamics. BEVs provides short-term flexibility that more effectively balances demand and supply fluctuations. In 2030, the V2G fleet provided all required balancing needs, rendering PHS inactive. By 2050, V2G operations reduced curtailment from 22 % in the *BAU* case to 5 %, enabling an increase in renewable electricity penetration from 55 % to 90 %. This demonstrates that smart mobility strategies can unlock latent system potential without requiring further capacity expansion.

In the *100 % REPS_{SM}* scenario, storage output reaches a maximum of 227 GWh. However, despite higher storage discharge, curtailment remains at 31 %, indicating that even with widespread V2G participation, additional grid flexibility measures are required, such as sector

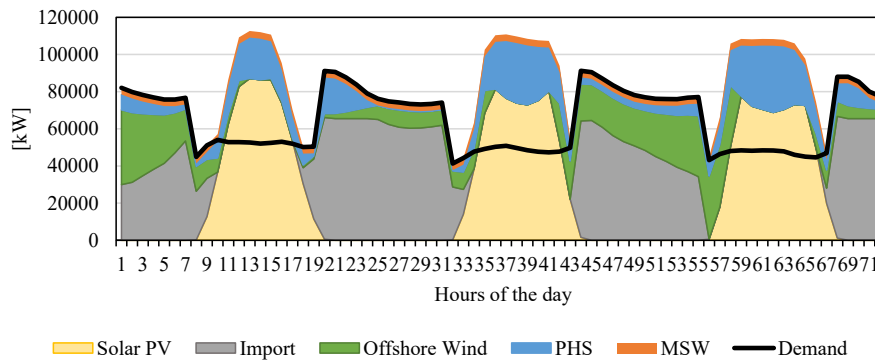


Fig. 3. Hourly electricity generation vs electricity demand in the 2050 BAU scenario. The sample corresponds to three days, from 03 to 05 September (sample hours 5905–5952).

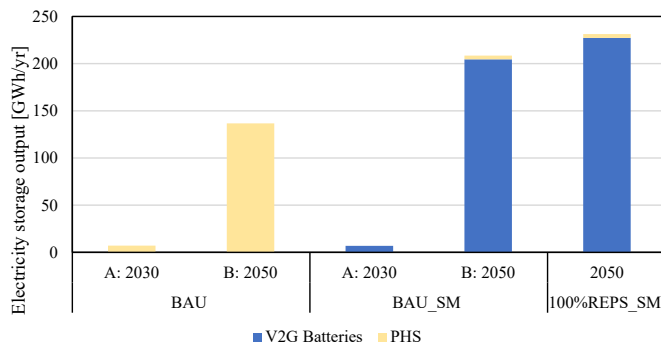


Fig. 4. Electricity storage output across all scenarios.

coupling with hydrogen production or higher storage capacity. Fig. 5 presents the storage dynamics in the 100 % REPS_{SM} scenario. V2G batteries exhibit a diurnal cycle, remaining in a state of full charge for most of the year, with notable disruption intervals in September, when RE generation is particularly low. This indicates that V2G primarily provides short-term balancing, while PHS shifts to a complementary role as seasonal backup to cover extended supply deficits. Bogdanov & Breyer [47] also confirm that co-utilizing EV batteries through smart charging and V2G consistently outperforms centralized storage for short-term flexibility, while longer duration storage options like PHS remain vital for seasonal balancing.

3.4. System costs and emissions

The total system costs across the scenarios are presented in Fig. 6. The annualised costs are broken down into investment, fixed operational, and variable costs. It can be seen that the total annualised system costs decline over time as the energy system transitions toward higher shares of RE and transport electrification. In the BAU scenario, system costs decrease from €105 million in 2030 to €83 million in 2050, primarily due to reduced fuel consumption as BEVs replace ICEVs. The BAU_{SM} scenario follows a similar trend but at a faster rate, with cost dropping from €108 million in 2030 to €65 million in 2050, driven by the additional flexibility provided by V2G services, which allow for more efficient use of RE resources. The 100 % REPS_{SM} scenario exhibits the lowest system costs, reaching approx. €62 million by 2050. This reduction is largely attributable to the elimination of fuel expenditures and the enhanced integration of RE, facilitated by smart charging and V2G storage solutions.

The cost structure in all the scenarios is dominated by variable costs, primarily driven by fuel expenditures for the transport sector. In the BAU scenario, fuel costs account for 86 % of total system costs in 2030,

decreasing to 64 % in 2050 as more vehicles transition to electric alternatives. In the BAU_{SM} scenarios, fuel costs initially remain high at 83 % in 2030 but drop substantially to 30 % by 2050. The 100 % REPS_{SM} scenario shows the greatest reduction in fuel dependency, with fuel expenditures constituting only 23 % of total costs in 2050, as the transport sector shifts 85 % to electricity.

The environmental benefits of this transition are equally significant. As illustrated in Fig. 7, energy-related CO₂ emissions are nearly eliminated in the 100 % REPS_{SM} scenario, with only a small residual amount coming from fossil fuel-dependant HDVs. In contrast, the BAU scenario shows CO₂ emissions increase from 519.8 kt in 2021 to 553.0 kt by 2030 and only a slight decline to 373.5 kt by 2050, due to the continued thermal PP operation. The BAU_{SM} scenarios achieve more significant CO₂ reductions through improved RE utilization via V2G integration. However, due to the limited capacity of PHS, the system still relies on fossil fuel, and a partial share of ICEVs in the transport sector prevents complete decarbonization. These results confirm that complete decarbonization of Santiago Island's energy system is only feasible through a fully renewable power sector, coupled with large-scale transport electrification and effective V2G integration.

3.5. Sensitivity analysis

A sensitivity analysis was carried out under the 2050 BAU_{SM} scenario to evaluate the impact of smart charging without V2G and the saturation effects of V2G. The results indicate that V2G significantly improves RE utilization, reducing curtailment and increasing the share of RE in the electricity mix. When only smart charging was considered, curtailed electricity reached 222.1 GWh, whereas V2G integration reduced to 95.98 GWh/year. As a result, the RE share in the electricity mix increased from 81 % to 96 %. These findings align with previous studies, such as Dominković et al. [54], which also demonstrated that smart charging alone enables higher RE penetration compared to BAU levels but is less effective than V2G in maximizing RE integration.

A second sensitivity analysis explored the impact of increasing the share of vehicles participating in V2G services. Raising the participation rate from 50 % to 75 % led to only a marginal RE share increase of 1 %. This suggests that beyond a certain threshold, additional V2G-enabled vehicles provided limited benefits. This outcome is consistent with findings from Child et al. [49], who observed similar saturation effects in V2G integration among marine vessels in Åland Islands.

4. Discussion

This study examines the feasibility of achieving 100 % RE penetration in Santiago Island's electricity mix based on existing government policies and RE potential. The results indicate that under the BAU scenario, even with the implementation of prioritized RE initiatives,

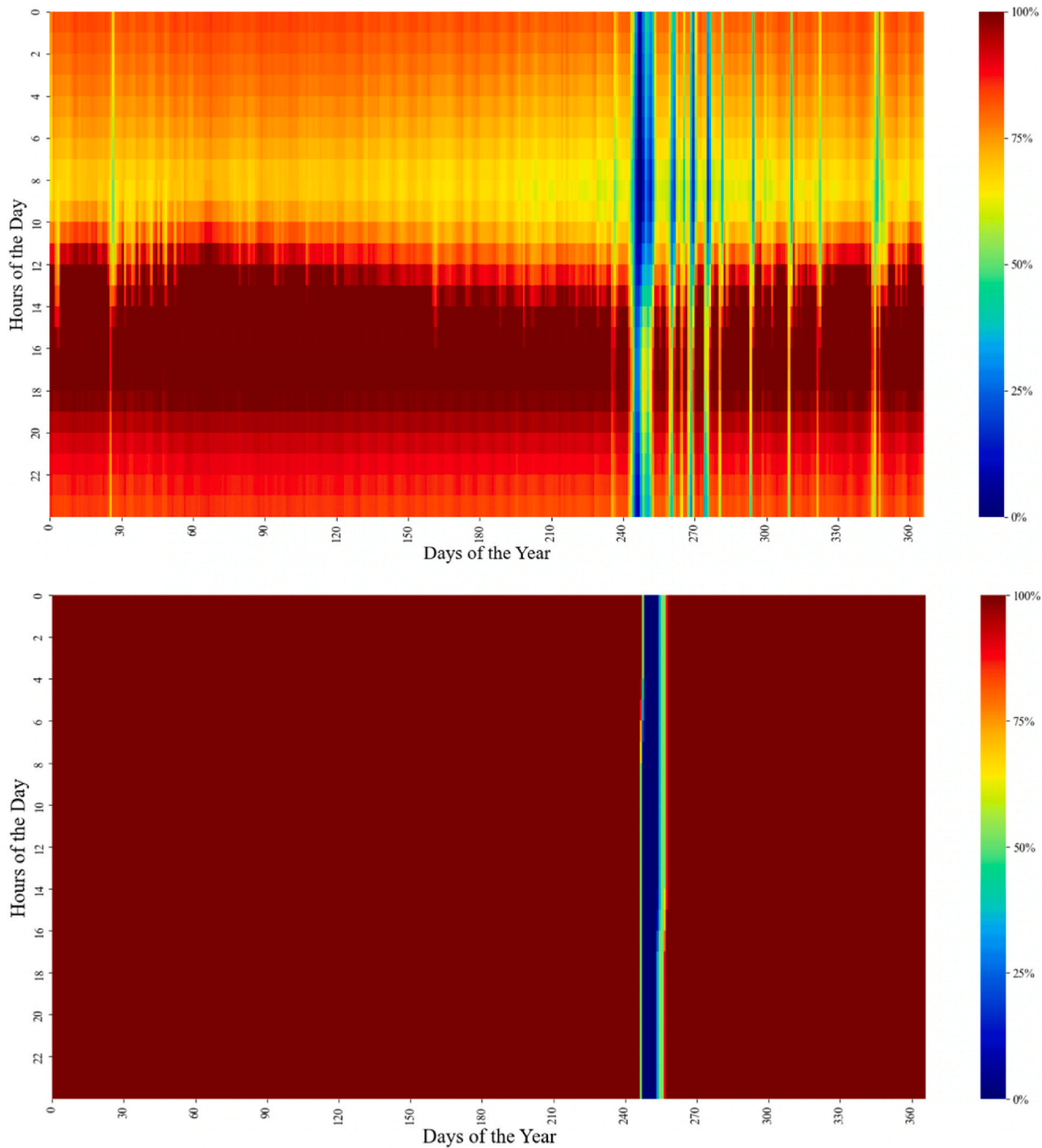


Fig. 5. Storage dynamics in the 100 % REPS_SM scenario, BEVs(top), PHS (below).

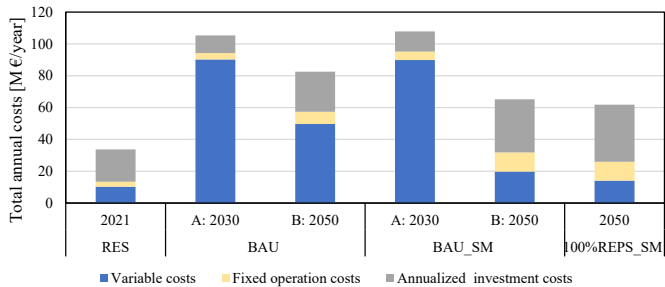


Fig. 6. Total annual cost of the energy system across the scenarios.

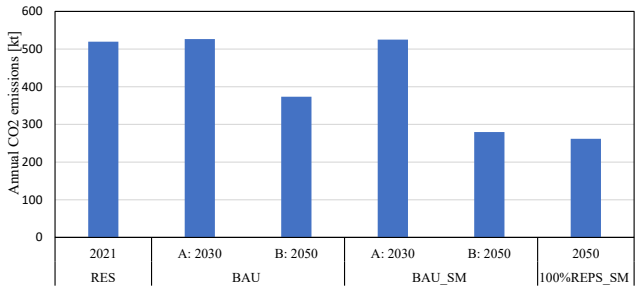


Fig. 7. Annual CO₂ emissions of the energy system across all scenarios.

achieving 100 % renewable electricity generation by 2050 is not technically feasible. The main challenge stems from the high seasonality of the intermittent nature of RE sources, particularly wind, which is further exacerbated by the increased electricity demand from EV charging and insufficient storage capacities. Achieving a high share of RE integration, therefore, requires adequate storage solutions to counteract intermittency issues and supply fluctuations. The validated hourly function of the EnergyPLAN model is, in this case, appropriate, allowing for a detailed assessment of supply-demand dynamics and flexibility requirements. Without effective energy storage mechanisms, fossil fuel-based generation remains necessary, limiting the full decarbonization potential of the energy system.

4.1. Electric vehicles with V2G batteries as a flexible solution in a highly renewable energy system

The impact of EV-induced electricity demand varies significantly across scenarios, particularly in how charging strategies influence RE integration. In the *BAU* scenario, dump charging occurs primarily during non-solar hours, forcing the system to rely on fossil fuel-based generation to meet both EV charging and electricity demand. This unregulated charging pattern not only strains conventional PP but also increases overall fuel consumption, highlighting not only the limitations of unmanaged EV adoption in a grid with high RE penetration and limited storage capacity, but it also underscores the importance of integrated energy planning that explicit couples the power and transport sector—an approach that forms the core of this study applied scenarios design. These findings are consistent with Pina et al. [55], who found that, even with high shares of renewables, EV charging can increase reliance on backup diesel generators, as seen on Flores Island.

In contrast, a smart charging strategy combined with V2G services, as demonstrated in the *BAU_{SM}* scenario, significantly enhances RE utilization, increasing by 38 % compared to the 2050 *BAU* scenario. The ability of V2G-enabled batteries to store and redistribute electricity eliminates curtailment in 2030 and enhances grid flexibility, ensuring that surplus RE is absorbed and discharged when needed. Xu et al. [56] similarly found that at V2G participation rates of 40–50 %, EVs can fully meet short-term grid storage requirements, confirming the effectiveness of EVs as distributed energy storage assets. However, without coordinated management, a high influx of EVs could lead to system imbalances and increased reliance on fossil fuels, as pointed out by Nornagoro et al. [57]. These risks highlight the importance of integrating smart charging with V2G services to enable bidirectional energy flow, allowing EVs to function not only as transportation assets but also as grid-supporting storage units [58].

The 100 % *REPS_{SM}* scenario further demonstrates that a fully RE-based system with high EV and V2G penetration is technically and economically feasible. Under the assumptions of this study, this system is 25.3 % more cost-effective than the *BAU* scenario, with total annual costs of €62 million, mainly due to significant reductions in fuel expenses and increased efficiency in RE utilization. These findings align with Hidaka et al. [59], who highlighted that leveraging the EV flexibility reduces fuel costs by improving load balancing and optimizing RE surplus absorption. In this scenario, V2G services contribute 20 % of total electricity generation, playing a pivotal role in balancing supply and demand throughout the year. This contribution is slightly higher than the 15.8 % reported by Dorotić et al. [5] for Korčula Island, suggesting that higher EV penetration and optimized charging strategies can further enhance system flexibility.

Finally, while V2G integration demonstrates opportunities for Santiago's energy transition, infrastructure constraints remain a major challenge. The island's grid is currently ill-equipped for bidirectional energy flows. Despite relatively high electricity access compared to other West African countries, frequent power outages and high distribution losses undermine grid reliability [28]. Scaling up EV adoption without improving grid stability could further strain the system,

especially during peak demand. As noted by Gicha et al. [60], such infrastructure limitations are common throughout Sub-Saharan Africa, alongside policy and regulatory framework gaps. In Santiago, regulatory frameworks are under consideration, but further research is needed on, e.g., business models for V2G participation. Future studies could assess whether private incentives, government-led, or utility-driven programs, such as managed charging, offer the most viable path. This would complement this research and contribute to Santiago's ambition to position itself as a regional leader in electric mobility among West African countries and other SIDS.

4.2. Electric mobility powered by renewables in African island nations

The integration of EVs and V2G services is necessary to achieve a fully decarbonized energy system on Santiago Island. However, its success depends on robust policy frameworks and an understanding of both technological and socio-economic factors. The transition to high shares of RE entails overcoming multiple barriers, including limited institutional capacity, inadequate planning frameworks to effectively strategize and secure investments [61], as well as outdated electricity infrastructure [62]. These barriers are particularly pronounced in SIDS, where remoteness, small populations, and logistical constraints further complicate the deployment of RE projects.

The long-term success of EV expansion in African islands also depends on the ability to address accessibility and affordability challenges. To address this, fiscal incentives such as tax exemptions and value-added tax (VAT) reductions, which have already been implemented in Santiago, play a significant role in making EVs more affordable and stimulating market adoption [63]. However, for low-income island nations like Cabo Verde, electrifying public transportation may represent a more viable and impactful first step toward decarbonization compared to stimulating private EV ownership. Research on the Philippines demonstrates that stronger financing support mechanisms, strategic deployment in short and flat routes, and investment in public charging infrastructure at terminals and other strategic locations are critical policy interventions for successful electric public transport adoption and offer better investment opportunities for transport operators and investors [64].

As demonstrated in this study, the adoption of V2G services offers a promising opportunity for increasing RE penetration and enhancing grid stability, but its implementation is contingent on both policy support and consumer willingness to participate. Yang et al. [65] emphasize that current incentive mechanisms are often inadequate and that concerns over battery degradation deter many EV owners from engaging in V2G services. However, the potential for generating revenue by selling power back to the grid could provide a strong financial incentive for participation [7,62]. Thus, effective policy frameworks should include well-structured incentives and public awareness campaigns to encourage EV owners to participate in V2G programs, thereby increasing grid resilience and RE integration [66].

Furthermore, transitioning away from petroleum imports could stabilize energy costs, improve trade balances, and create opportunities for reinvesting cost savings into EV infrastructure, particularly in public charging networks and public transportation systems (Cantarero, 2019). These investments not only accelerate EV adoption but also stimulate job creation in sectors such as renewable energy, transportation, and infrastructure development (Cantarero, 2019). According to IRENA [2], expanding wind and solar projects in African SIDS could create more than 3000 jobs by 2030, demonstrating the potential of clean energy

investments in driving sustainable development.

Given that African SIDS generally have higher literacy rates¹ and smaller populations compared to mainland countries in Africa, targeted awareness campaigns could play a key role in promoting EV and V2G adoption. By leveraging existing education levels and policy experiences, these island nations could accelerate the transition to sustainable transport faster than mainland African regions. The policy-driven scenarios modelled in this study reflect Santiago Island's real decarbonization targets and regulation contexts, thus enhancing the relevance of the results for decision-makers. With appropriate policy reforms, targeted research, and well-designed financial mechanisms, V2G adoption could become a scalable and replicable solution, demonstrating its feasibility in real-world applications [67].

4.3. Limitations

The simulations presented in this paper offer valuable insights into the integration of EVs and V2G technologies in Santiago Island's power system, assuming a high-RE penetration level driven by local policies. However, several limitations are acknowledged.

First, the modelling assumes the same demand curve shape without capturing the possible structural changes over all the modelled years and scenarios. Second, the study assumes that the existing grid infrastructure can accommodate the implications of integrating high shares of variable renewables, EV charging, and V2G functionality, without additional expansion. This is an important limitation, particularly in the context of SIDS systems, as the transmission and distribution network's capacity to manage fluctuating generation and demand, as well as bidirectional power flows, may present technical and economic challenges affecting the feasibility of large-scale RE penetration and V2G participation. Third, the PHS is modelled as an aggregated capacity with simplified operation, assuming ideal round-trip efficiency and availability, without accounting for constraints such as ramping limitations or reservoir design. Fourth, while this study currently focuses on private BEVs, broader assessments should also include other transport modes such as freight, public transit, maritime, and aviation sectors to evaluate their cumulative system impacts. Finally, battery degradation and associated lifecycle costs are not considered, although they can influence both V2G participation and system-level economics over time.

Future studies should incorporate more detailed grid assessments, account for the variability and evolving patterns resulting from structural changes in demand over the years, and explore demand-side management strategies to improve load flexibility. In addition, exploring the inter-island grid connection within the Cabo Verde archipelago could help alleviate these network constraints by providing additional flexibility and resilience at the system level. This option warrants deeper investigation as an alternative pathway to strengthen grid adequacy for high renewables and smart mobility integration. Long-duration storage options such as power-to-X should also be further analysed to address seasonal mismatches that cannot be solved by short-term V2G flexibility alone. Finally, integrating carbon pricing, battery degradation, and externalities costs would enhance the economic realism and policy relevance of future modelling efforts.

5. Conclusions

This study presents an hourly-resolution based scenario modelling of electric vehicles (EVs) integration and vehicle-to-grid (V2G) services within the energy system of an African Small Island Developing State

¹ Seychelles, São Tomé and Príncipe, Mauritius, and Cabo Verde are ranked among the highest African countries in terms of literacy rates between 2018 and 2022 (Statista, 2024). Individuals with higher income and education levels are more prone to adopt EVs due to higher purchasing power and environmental awareness [68].

(SIDS), using Santiago Island (Cabo Verde) as a case study, addressing geographical and methodological research gaps from the literature. By modelling three scenarios in EnergyPLAN to align with national decarbonization targets for 2030 and 2050, this research addresses a key gap in the literature on transport electrification and its role in enhancing renewable energy (RE) integration in African island contexts.

Among the modelled scenarios, the 100 % REPS_SM scenario emerges as the most technically and economically effective pathway, achieving the highest RE penetration, the lowest total annual costs, and near-zero emissions. The results demonstrate that smart charging and V2G-enabled battery electric vehicles (BEVs) can effectively function as distributed energy storage systems, enhancing synergies between the power and transport sectors. Even in the less ambitious BAU_SM scenario, V2G services enable Santiago to meet all short-term storage needs by 2030, demonstrating the potential of EVs to support grid flexibility and RE integration at minimal additional infrastructure cost.

However, successful V2G implementation depends not only on technical feasibility but also on robust policy frameworks that include appropriate incentives, clear regulations, and strategic investment planning. For Santiago and similar SIDS, such frameworks must account for infrastructure limitations and the high cost of imported fuels, suggesting that V2G could serve as a supportive mechanism for improving energy security and resilience, especially in systems limited by imported fuels and limited flexibility alternatives.

Despite the added flexibility from EVs and V2G services, the results highlighted that these solutions alone are insufficient to fully address seasonal supply-demand mismatches. In the 100 %REPS_SM scenario, curtailment remains at 31 %, even with high V2G participation and pumped hydro storage (PHS). This indicates the need for complementary seasonal storage solutions, such as power-to-X technologies, to convert surplus electricity into storable energy carriers, thereby reducing reliance on fossil fuel backup.

The case of Santiago Island demonstrates how integrated planning across the power and transport sectors, anchored in smart charging and V2G technologies, can advance the feasibility of 100 % renewable penetration in African island settings. With coordinated policy support and community engagement, African islands can become trailblazers in climate action, providing replicable models for sustainable transport and energy system integration across Sub-Saharan Africa and other island regions.

CRedit authorship contribution statement

Katia D.S. Gomes: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Richa Adhikari:** Writing – review & editing, Validation, Software, Methodology, Formal analysis, Conceptualization. **Maria Movsessian:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2025.137868>.

Data availability

Data will be made available on request.

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